8. Input and output signals – External control (PLC)

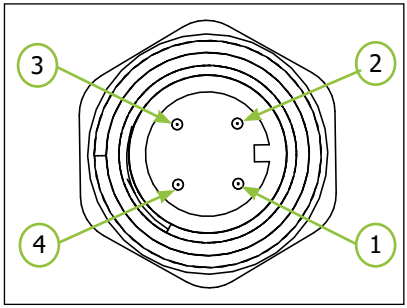
To command the LUB-V via an external control (PLC), it is necessary to switch the LUB-V to pulse mode in the Settings menu (chap. 6.3.4).
The LUB-V operates in pulse mode as a pulse-controlled lubrication system only if unalterable control signals of defined length are transmitted from the PLC to the LUB-V via PIN 2. The LUB-V signals the respective condition to the PLC via high/low levels, which must be tapped at PIN 4, and thus enables comprehensive control of the various conditions. For the integration of the LUB-V into an external control system, one input and one output each must be provided on the control side.

DANGER

Machine elements that are not lubricated can cause failures resulting in serious injury or death.

- To operate the LUB-V via an external controller (PLC) in pulse mode, a program corresponding to the communication protocol must be created in the PLC.
- After sending a control signal, the response signal of the LUB-V must be waited for, evaluated and interpreted in order to prevent an uncontrolled failure of the LUB-V.

8.1 Pin assignment – External control (PLC)



PIN-Assignment-external control (PLC):

PIN	Assignment	Colour
1	+24 V DC	brown
2	Input Signal PLC → LUB-V	white
3	Ground	blue
4	Output Signal LUB-V → PLC	black

Type: M12x1 female connector; 4-pin, A-coded

For the electrical connection to the external control system (PLC) of a machine, the LUB-V has a 4-pin interface, which is designed as a plug-in connection with the industry-standard M12x1 connection.

ⓘ The LUB-V can be switched off completely in pulse mode by switching off the supply voltage. The settings made are not lost in the process. After the supply voltage is reconnected, the LUB-V checks itself automatically and does not operate until it receives a control signal from the PLC.

ⓘ The output signal at PIN 4 must be tapped at an external control for further processing. The maximum permissible output current must not exceed I_{max} < 20mA. No inductive load (e.g. relay) may be connected!

ⓘ After a longer standstill of the LUB-V, it is recommended to trigger the control signal „1 lubrication stroke" twice manually (100 ms) (chap. 8.2.1).

8.2 Input signals – External control (PLC)

The LUB-V provides the following invariably defined control signals (input signals), which must be transmitted from the PLC to the LUB-V via PIN 2 of the electrical M12x1 interface as high level (+24 V DC).
The control signals must be generated as high level (+24 V DC) by the external control (PLC) over specific times, each with a tolerance of ±25 milliseconds.

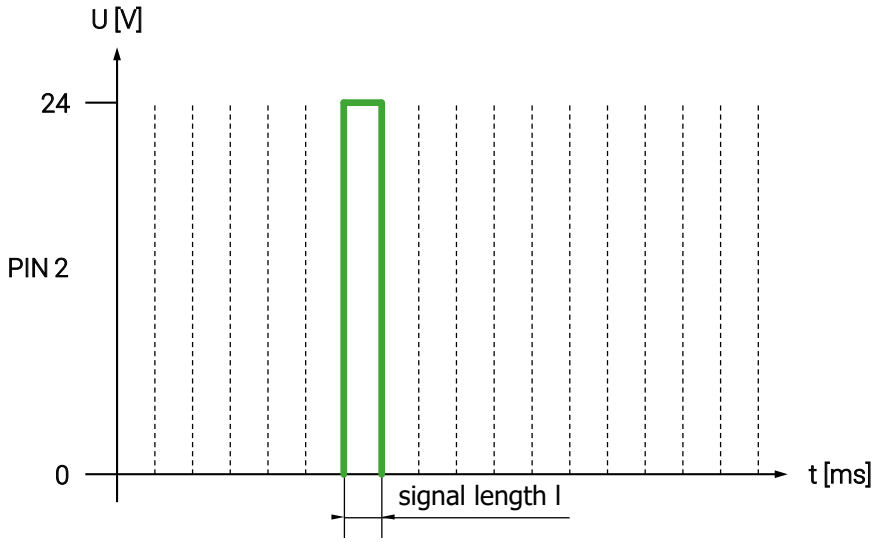
The LUB-V processes the following control/input signals:

Signal length I [ms]	Function	Detail
100	1 lubrication stroke	Chap. 8.2.1
900	Filling	Chap. 8.2.2
1000	Cancel filling	Chap. 8.2.3
1600	Status request (sign of presence)	Chap. 8.2.4
1700	Acknowledge error	Chap. 8.2.5

ⓘ The LUB-V in pulse mode only processes the control signals listed in the table up to a maximum length of 1700 ±25 milliseconds. If a high level (+24 V DC) is present outside the tolerances, the LUB-V returns a message for an impermissible control signal as a response, see Chap. 8.3. If a high level (+24 V DC) is present at PIN 2 of the electrical interface for longer than 1725 milliseconds, the LUB-V also returns a message for an impermissible control signal as a response, see Chap. 8.3.

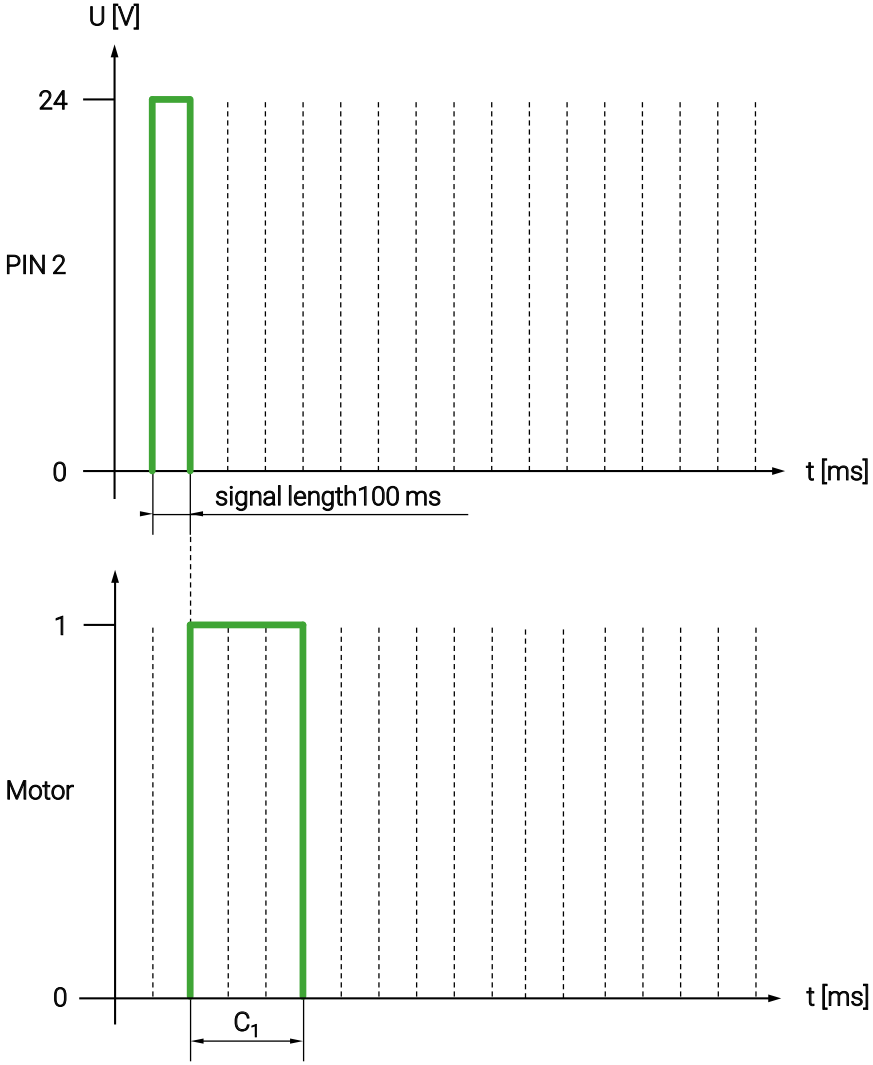
ⓘ LUB-V continues to process the signal lengths used by devices between 2010 and 2022! (2 seconds: 1 stroke, 12 seconds: Filling function, 14 seconds: Acknowledge error) The length of the first signal received determines which control signals it reacts to. A changeover is made by interrupting the voltage for a few seconds.

ⓘ The following diagram shows the general characteristics of a control signal that must be generated by the PLC or external controller and sent to the LUB-V controller via PIN 2.



8.2.1 Control/input signal „1 lubrication stroke"

The control signal „1 lubrication stroke" triggers a single lubrication stroke.



Description:

✓ The LUB-V is properly connected to an external controller via the electrical interface and connected to the power supply.
✓ Pulse mode is activated on the LUB-V.
✓ There are no errors on the LUB-V; the LUB-V is ready for operation.

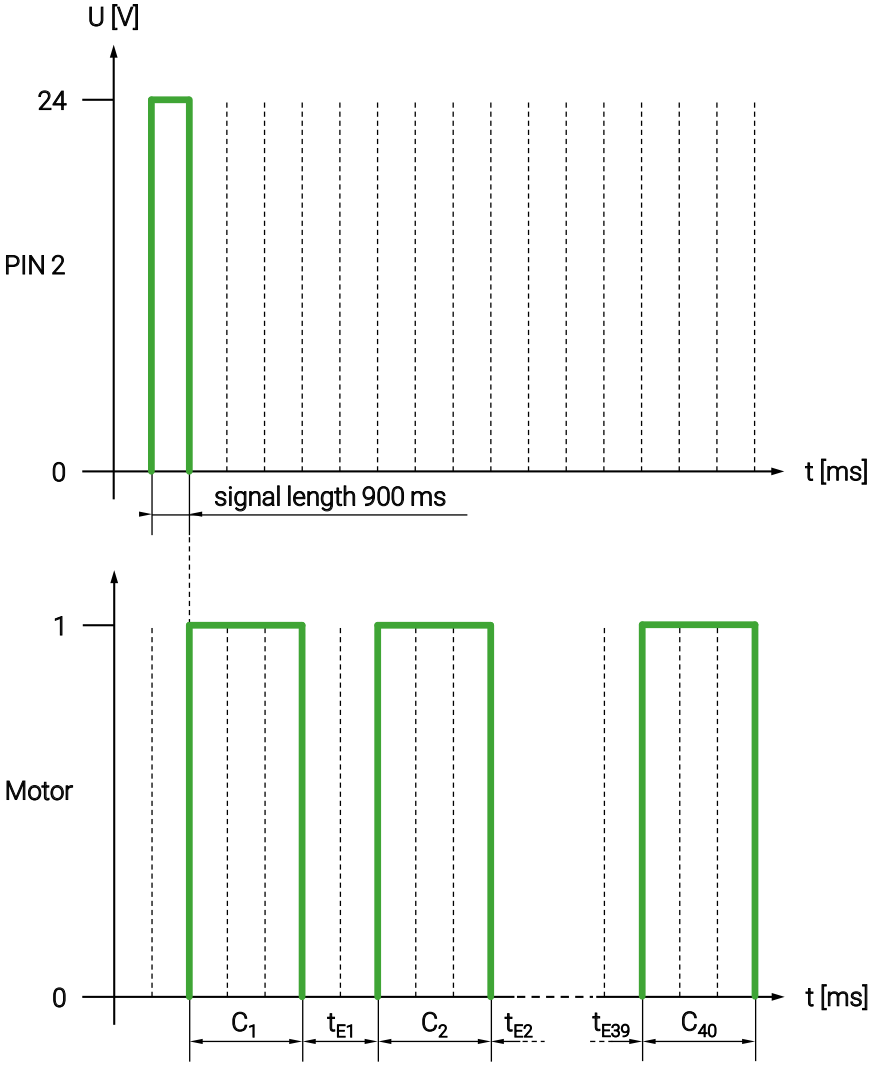
● The LUB-V sends a permanent output signal (high level) to PIN 4, which indicates readiness for operation to the external controller (PLC). This output signal must be present permanently and uninterruptedly for >500 milliseconds. Only if this prerequisite is fulfilled, an activation by the external control is possible at all.
● The control signal „1 lubrication stroke" with signal length of 100 (75 ... 125) milliseconds high level can be sent from the external control (PLC) to the LUB-V.
● While a high level is present at PIN 2 of the LUB-V as an input signal from the external controller (PLC), the symbol indicating the pulse mode flashes in the display for the length of the signal.
● Immediately after the control signal drops, the lubrication stroke C of the LUB-V starts and 0.15 ml of lubricant is delivered to the outlet. The LUB-V monitors this lubrication stroke for the entire duration.
● The lubrication stroke time depends on various conditions, including the counterpressure present or built up in the hydraulic system and the temperature. For the LUB-V, the lubrication stroke duration is 7...17 seconds (t_C = 7...17 seconds).
● During the lubrication stroke C, a numerical value 1...70 is shown in the display, which indicates the approximate back pressure in bar. The back pressure is displayed in relation to the outlet.
● At the end of a lubrication stroke C, the LUB-V sends an output signal to PIN 4, which provides information about the past lubrication stroke as well as other states of the LUB-V for evaluation at a PLC or other external controller (see chapter 8.3).

ⓘ At the earliest >500 milliseconds after the end of the response signal, a possible next control signal can be sent from the external control (PLC). For the characteristics and interpretation of the output signal, refer to Chap. 8.3. In the meantime, the LUB-V does not process any control signals.

ⓘ If the integrated microelectronic of the LUB-V has detected an error during or immediately after the end of the lubrication stroke, it will be transmitted to the external control (PLC) by the corresponding output signal (chap. 8.3).

8.2.2 Control/input signal „Filling"

The „Filling" control signal triggers the filling function by the external control. A total of 40 lubrication strokes are performed automatically one after the other.



Description:

✓ The LUB-V is properly connected to an external controller via the electrical interface and connected to the power supply.
✓ Pulse mode is activated on the LUB-V.
✓ There are no errors on the LUB-V; the LUB-V is ready for operation.

● The LUB-V sends a permanent output signal (high level) to PIN 4, which indicates readiness for operation to the external controller (PLC). This output signal must be present permanently and uninterruptedly for >500 milliseconds. Only if this prerequisite is fulfilled, an activation by the external control is possible at all.
● The control signal „Filling" with signal length of 900 (875 ... 925) milliseconds high level can be sent to the LUB-V from the external control (PLC).
● While a high level is present at PIN 2 of the LUB-V as an input signal from the external controller (PLC), the symbol indicating the pulse mode flashes in the display for the length of the signal.
● Immediately after the control signal drops, the lubrication stroke C of the LUB-V starts and 0.15 ml of lubricant is delivered to the outlet. The LUB-V monitors this lubrication stroke for the entire duration.
● The lubrication stroke time depends on various conditions, including the counterpressure present or built up in the hydraulic system and the temperature. For the LUB-V, the lubrication stroke duration is 7...17 seconds (t_C = 7...17 seconds).
● During the lubrication stroke C, a numerical value 1...70 is shown in the display, which indicates the approximate back pressure in bar. The back pressure is displayed in relation to the outlet.
● At the end of a lubrication stroke C, the LUB-V sends an output signal to PIN 4, which provides information about the past lubrication stroke as well as other states of the LUB-V for evaluation at a PLC or other external controller (see chapter 8.3).
● A total of 40 lubrication strokes C and 39 relaxation phases t_E of two seconds length take place in immediate succession between the lubrication strokes. In the process, 40 x 0.15 ml = 6.0 ml of lubricant are delivered from the cartridge to the outlet.

ⓘ At the earliest >500 milliseconds after the end of the last output signal, a possible next control signal can be sent from the external control (PLC). During the filling function, the control signal „Cancel filling" (chap. 8.2.3) is the only control signal that the LUB-V can process.

ⓘ If the integrated microelectronic of the LUB-V has detected an error during or immediately after the end of the lubrication stroke, it will be transmitted to the external control (PLC) by the corresponding output signal (chap. 8.3).
ⓘ The filling function can be stopped with the „Cancel filling" control signal, see section 8.2.3.

8.2.3 Control/input signal „Cancel filling"

The „Cancel filling" control signal can be used to stop the filling function activated by the „Filling" control signal. During the filling function, the „Cancel filling" control signal is the only control signal that the LUB-V can process.

Description:

✓ The LUB-V is properly connected to an external controller via the electrical interface and connected to the power supply.
✓ Pulse mode is activated on the LUB-V.
✓ The filling function has been activated by the „Filling" control signal.

● The LUB-V carries out the filling function.
● The control signal „Cancel filling" with a signal length of 1000 (975 ... 1025) milliseconds high level can be sent from the external control (PLC) to the LUB-V during the relaxation time t_E between two lubrication strokes C.
● At the end of the transmission of the „Cancel filling" control signal, the LUB-V stops the filling function directly. If the „Cancel filling" control signal is sent during a lubrication stroke, the current lubrication stroke is completed and then the filling function is canceled.

ⓘ At the earliest >500 milliseconds after the end of the response signal, a possible next control signal can be sent from the external control (PLC). For the characteristics and interpretation of the output signal, refer to Chap. 8.3. In the meantime, the LUB-V does not process any control signals.

8.2.4 Control/input signal „Status request"

The „Status request (sign of presence)" control signal can be used to check the last status of the LUB-V and the last lubrication stroke. In addition, the „Status request (sign of presence)" control signal can be used to cyclically check whether the LUB-V can be reached.

Description:

✓ The LUB-V is properly connected to an external controller via the electrical interface and connected to the power supply.
✓ Pulse mode is activated on the LUB-V.

● A high level is applied to PIN 4 of LUB-V for >500 milliseconds.
● The control signal „Status request (sign of presence)" with signal length of 1600 (1575 ... 1625) milliseconds high level can be sent from the external control (PLC) to the LUB-V.
● After the end of the transmission of the control signal „Status request (sign of presence)", the LUB-V repeats the output signal of the past lubrication stroke on PIN 4.

ⓘ At the earliest >500 milliseconds after the end of the response signal, a possible next control signal can be sent from the external control (PLC). For the characteristics and interpretation of the output signal, refer to Chap. 8.3. In the meantime, the LUB-V does not process any control signals.

8.2.5 Control/input signal „Acknowledge error"

The „Acknowledge error" control signal is used to acknowledge the E2 overpressure or E3 over/undervoltage errors after they have occurred previously and have been communicated by the LUB-V to the PLC. It is the only control signal that the LUB-V can process if an error message was sent in response to the previous motor run. Regardless of the principle possibility of remote acknowledgement of an error, it is essential to identify and eliminate the cause when an error message is present.

Description:

✓ The LUB-V is properly connected to an external controller via the electrical interface and connected to the power supply.
✓ Pulse mode is activated on the LUB-V.
✓ There is an E2 or E3 error at LUB-V.

● The output signal sent from the LUB-V to PIN 4 to the external controller reports an error message E2 or E3 back.
● An identification of the cause of the error message has taken place on the LUB-V and it has been eliminated.
● The control signal „Acknowledge error" with signal length of 1700 (1675 ... 1725) milliseconds high level can be sent from the external control (PLC) to the LUB-V.
● After the end of the control signal, the integrated microelectronic checks the LUB-V independently:
➲ If this internal check is **successful**, the error E2 or E3 is thus acknowledged and the LUB-V is ready for operation again.
➲ If this internal check is **not successful**, the LUB-V continues to send an output signal that defines an error. The error is still present. In this case, check the lubrication point and the LUB-V again and then send the „Acknowledge error" control signal again. If the second check is again unsuccessful, disassemble the LUB-V with the lubricant cartridge screwed on and return the complete lubrication system to the manufacturer with a detailed description of the error.

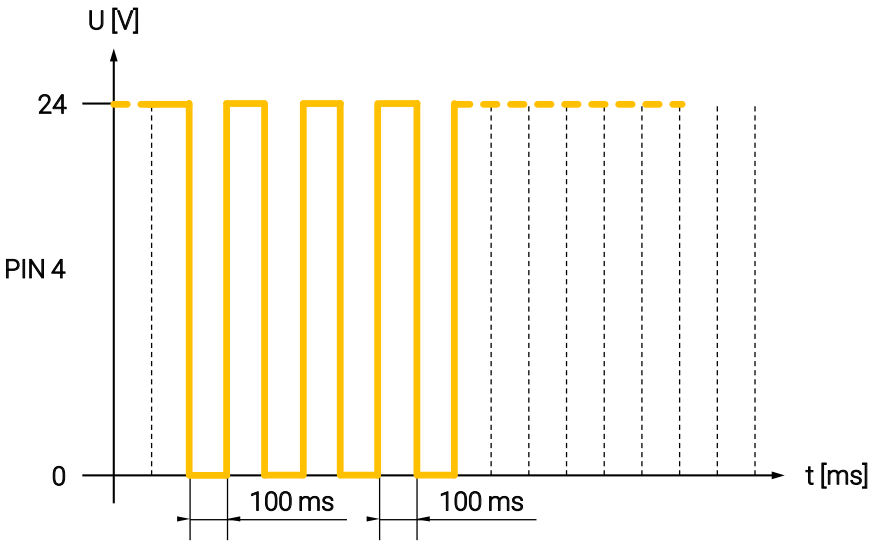
ⓘ ⓘ At the earliest >500 milliseconds after the end of the response signal, a possible next control signal can be sent from the external control (PLC). For the characteristics and interpretation of the output signal, refer to Chap. 8.3. In the meantime, the LUB-V does not process any control signals.

8.3 Output signals – External control (PLC)

The LUB-V has invariably defined output signals that are transmitted to the PLC via PIN 4 of the M12x1 electrical interface depending on the condition of the LUB-V.
Each time a control signal is sent from the PLC to the LUB-V, the LUB-V sends an output signal after the action caused by the control signal has been executed, which must be waited for, evaluated and interpreted by a suitable PLC program.

A high level (+ 24 V DC) is permanently present at PIN 4. During the transmission of the output signal, the signaling for information transmission changes several times from a high level to a low level and back again. After sending an output signal, a high level is permanently present at PIN 4.

The following graphic shows an example of an output signal.



Information about the status of the LUB-V and about the past motor run must be evaluated via an evaluation of the number of edge changes of the output signal. For the evaluation of the edge changes, the rising edge (change of the output voltage at PIN 4 of the LUB-V from a low level (0 V) to a high level (+24 V DC)) must always be counted. The counted number of edge changes corresponds to a one-to-one state of the LUB-V.
The output signal is sent with a frequency of f = 5 Hz. The low level is a voltage of + 0 to 5 V DC, the high level is a voltage between + 17 and + 27 V DC.

The following table shows all the output signals that are sent from the LUB-V to the external controller depending on its status.

Number of edge changes	Information	Remedy
1	Filling function canceled	none necessary, purely informative
2	Past lubrication stroke OK	none necessary, purely informative
3	Past lubrication stroke OK, cartridge soon empty	Buy a new cartridge in time.
4	For past lubrication stroke Overpressure (E2) at outlet 1	▶ Check the lubrication point and eliminate the cause ▶ Acknowledge error (Chap. 8.2.5)
5	With past lubrication stroke Overpressure (E2) at outlet 2 (if present)	▶ Check the lubrication point and eliminate the cause ▶ Acknowledge error (Chap. 8.2.5)
12	Cartridge empty	▶ Change the cartridge (see chap. 9.2) ⓘ No need to acknowledge the error; it is automatically cleared after the shutdown action.
14	Over-/undervoltage (E3)	▶ Check the power supply of the LUB-V. ▶ Acknowledge error (Chap. 8.2.5)
15	Internal device error (E4)	▶ Disassemble the LUB-V and return it to the manufacturer together with the lubricant cartridge and a description of the fault.
16	Inadmissible, undefined control signal received	▶ Check the PLC program for operating the LUB-V for correctness.

	DANGER
	Machine elements that are not lubricated can cause failures resulting in serious injury or death. ● After sending a control signal, the response signal of the LUB-V must be waited for, evaluated and interpreted in order to prevent an uncontrolled failure of the LUB-V.

ⓘ If there is a low level at PIN 4 in pulse mode, but the operating voltage is connected, check the connection cable from external control to PIN 4 of the LUB-V. If you cannot detect a problem during the check, a serious device error has occurred that cannot be remedied by you on site. In this case, disassemble the LUB-V with the lubricant cartridge screwed on and return the complete lubrication system to the manufacturer with a detailed description of the fault.

ⓘ The LUB-V always sends a specific output signal within 30 sec. after sending an arbitrary control signal. If this is not the case and the LUB-V no longer responds to any control signal, a time-out has been detected. To check the communication readiness of the LUB-V in this case, send the control signal „Status request (sign of presence)". If again >30 sec. pass without a response signal from the LUB-V, disconnect the supply voltage from the LUB-V for at least 10 sec. and send the control signal „Status request (sign of presence)" again after reconnecting the supply voltage. If communication with the LUB-V is still not possible, disassemble the LUB-V with the lubricant cartridge screwed on and return the complete lubrication system to the manufacturer with a detailed description of the error.